

ModelSmith - Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR-01	Performance	Interactive API operations should remain responsive; long-running analysis, optimization and benchmarking jobs shall run asynchronously.
NFR-02	Scalability	The architecture shall support multiple users and concurrent background jobs, with independently scalable API and worker services.
NFR-03	Reliability	Job state shall be persisted; recoverable failures shall support retry, and incomplete artifacts shall not be marked successful.
NFR-04	Availability	Critical services shall expose health checks and use appropriate backup/recovery procedures for persistent data.
NFR-05	Security	Protected operations shall require authentication and authorization; least privilege shall be applied across users, services and storage.
NFR-06	Data Protection	Models, user data and artifacts shall be encrypted in transit and at rest and isolated between projects/users.
NFR-07	Model Security	Uploaded models shall undergo file validation/security checks and execute in isolated, resource-controlled worker environments.
NFR-08	Data Integrity	Models, versions, plans, predictions, jobs, benchmarks and artifacts shall remain traceably linked.
NFR-09	Reproducibility	An optimization result shall record the model version, plan, hardware/runtime context and execution metadata required to reproduce it.
NFR-10	Usability	The workflow shall be clear and structured; recommendations shall explain expected benefits and important trade-offs.
NFR-11	Compatibility	The platform shall support documented model formats and target hardware/runtime combinations through modular adapters.
NFR-12	Maintainability	Analysis, planning, prediction, execution and export components shall be modular and independently extensible.
NFR-13	Observability	The system shall provide structured logs, job status, health/resource metrics and actionable failure information.
NFR-14	Privacy	Access to project resources shall be restricted to authorized users and retention/deletion rules shall be defined for models, logs and artifacts.

Key Quality Priorities

Security • Reliability • Performance • Scalability • Explainability • Reproducibility • Maintainability
• Usability • Observability